FORM HS-04 · REV. 4 POST-STASIS REFRESHER · MODULE 4 OF 9 POST AT EVERY TOWER BASE AND SCAFFOLD

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Health & Safety

TIE OFF.

FALLS KILL MORE CONSTRUCTION WORKERS THAN ANYTHING ELSE. A CITY MEANS TOWERS. EVERY DECK ABOVE ZERO IS A FALL.

THE 6-FOOT RULE

AT 6 FEET, YOU NEED FALL PROTECTION.

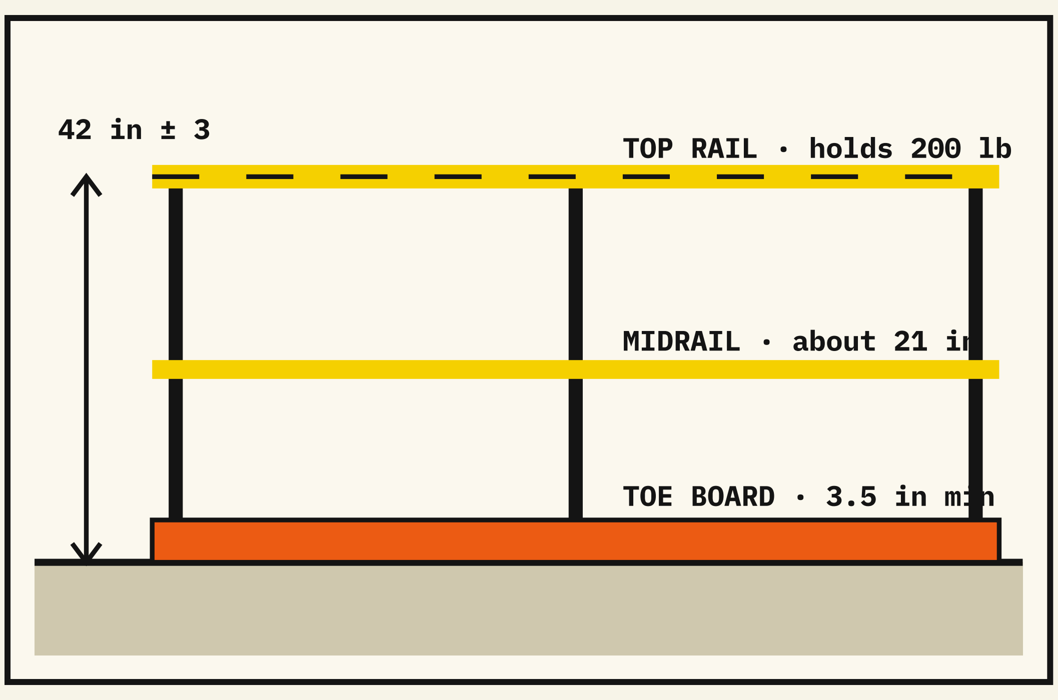
Any edge, hole or opening where you could fall **6 feet or more** to a lower level needs a guardrail, a safety net, or a personal fall arrest system. Scaffolds start at **10 feet**. On this site, Deck 1 and above always counts.

6 ft FALL PROTECTION REQUIRED	**10 ft** SCAFFOLD FALL PROTECTION	**#1** KILLER IN CONSTRUCTION

Just revived? Fail the balance check and you stay at Deck 0 until Med Bay clears you.

GUARDRAILS

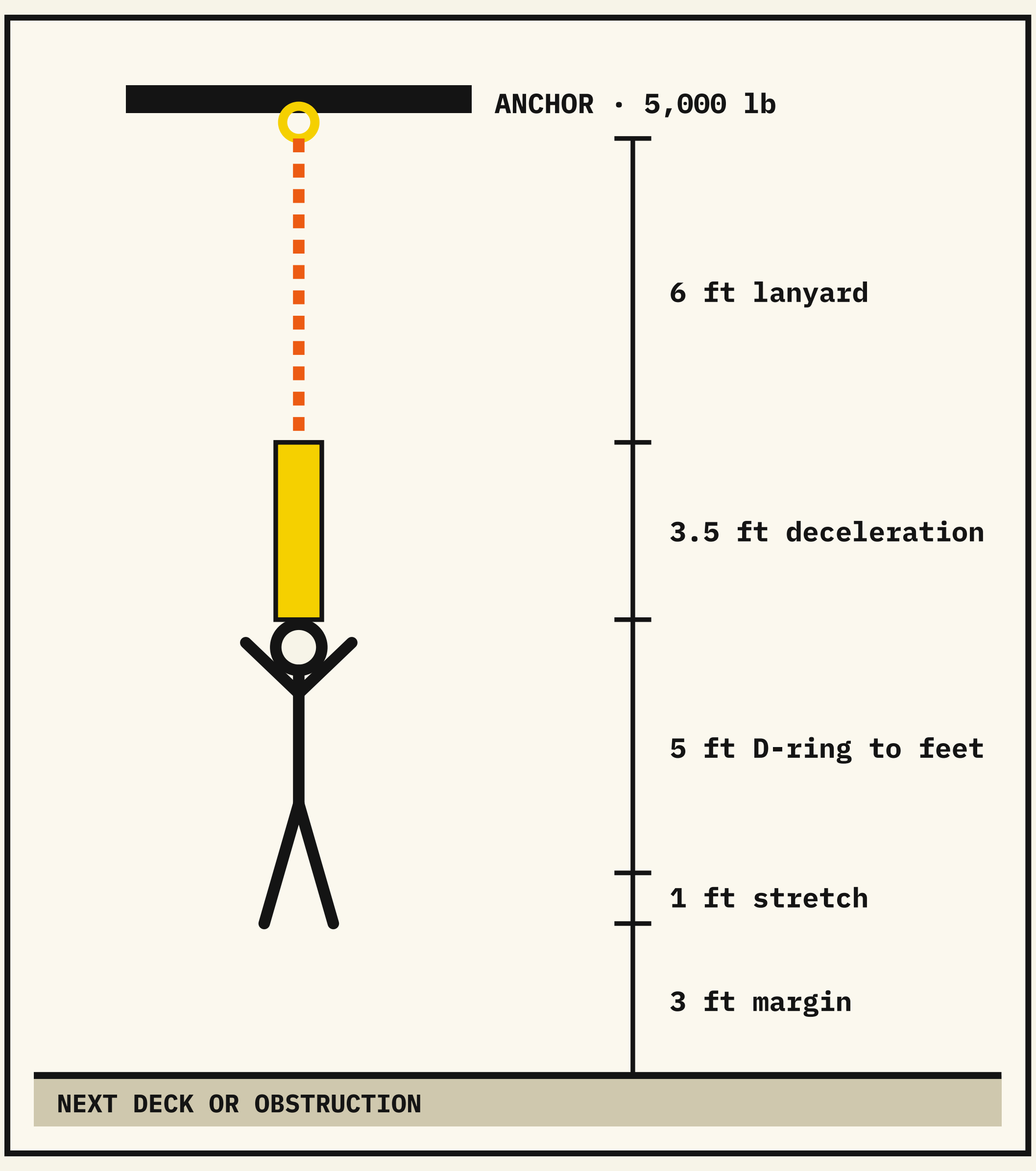
FIRST CHOICE. PROTECTS EVERYONE, EVERY TIME.



- **Top rail 42 inches** above the walking surface, plus or minus 3 inches.
- Top rail must hold **200 lb** of force pushed outward or down without failing.
- **Midrail about halfway** (around 21 inches), or mesh or screen that fills the gap.
- **Toe board at least 3.5 inches** tall wherever tools or material can be kicked off the edge.
- Removed a section to land a load? **Tie off** until it's back. Then put it back.
- Never lean on, sit on or climb a guardrail. Never hang a load from one.

FALL CLEARANCE

KNOW HOW FAR YOU'LL DROP BEFORE YOU CLIP IN



18.5 ft
CLEARANCE NEEDED BELOW THE ANCHOR

- **Free fall: 6 feet max.** Anchor at or above your D-ring so the lanyard can't let you drop farther.
- **Deceleration: up to 3.5 feet** while the shock absorber tears open.
- Add your body below the D-ring (about 5 ft), harness stretch (about 1 ft) and a 3 ft margin.
- **Less than 18.5 ft below?** A 6 ft shock-absorbing lanyard will hit the deck or steel before it stops you. Use a self-retracting lifeline (SRL) instead.
- **Anchors:** 5,000 lb per worker attached, or designed by a qualified engineer with a 2:1 safety factor. Never anchor to guardrails, conduit, pipe or a load.
- Swing fall: an anchor off to the side swings you into structure. Anchor overhead.

FALL ARRESTED

Hanging in a harness cuts off blood flow from the legs. **Suspension trauma** can make a conscious worker pass out in minutes and can kill.

NO RESCUE PLAN, NO CLIMB.

Your crew lead names the rescue method and the rescue gear before anyone goes up. Waiting for Fire & Rescue to arrive from the Hub takes too long.

1. **Ch 1: "Emergency, emergency, emergency. Fall, worker suspended"**, with sector-grid and deck.
2. **Talk to them.** Keep them calm and moving their legs. Have them stand in their relief straps or push against the structure.
3. **Run the rescue plan now:** rescue ladder, lift, rescue descent device or haul system. Don't wait for medics.
4. **Once down, lay them flat.** Treat as a trauma patient. Watch breathing.
5. Take the harness and lanyard out of service. Tag it. It goes to Safety for inspection.

HARNESS CHECK, EVERY USE

- **Webbing:** no cuts, frays, burns, weld spatter holes, chemical stains or hard, glazed spots.
- **Stitching:** no pulled or broken threads.
- **Hardware:** D-rings and buckles not bent, cracked or pitted. Snap hooks lock and close by themselves.
- **Lanyard:** shock-absorber pack intact. Torn pack or red indicator showing = it already caught a fall.
- **Fit:** chest strap mid-chest, leg straps snug enough for a flat hand under them, D-ring between your shoulder blades.
- Any doubt: tag it out. Draw a new one from the Quartermaster.

LADDERS

4:1 UP 4, OUT 1	**3 ft** ABOVE LANDING	**3** POINTS OF CONTACT

- Set the base 1 foot out for every 4 feet up.
- Extend the rails 3 feet above the landing, and tie the top off.
- Always 2 hands and a foot, or 2 feet and a hand. Carry tools on your belt or haul them up.
- Face the ladder going up and down. Never stand on the top two rungs of a stepladder.
- Metal ladders conduct and get hot. Fiberglass near power. Gloves on sun-baked rungs.

HOLES

- Every floor hole gets a **cover or a guardrail**.
- Covers hold **twice** the heaviest load that could cross them: workers, equipment, material.
- Secured so wind, equipment or feet can't shift them.
- Marked **"HOLE"** or **"COVER"** in paint or on the cover.
- Never lift a cover you didn't set. Tell your lead.

DROPPED OBJECTS

- A wrench dropped from Deck 10 can kill someone at Deck 0.
- **Tool lanyards** on every hand tool used at height.
- Toe boards and netting on open edges.
- **Exclusion zone** roped off below overhead work. Nobody walks through it.
- Never throw anything down. Lower it.

WIND LIMITS

Site rule. Ch 16 Atmospheric posts wind at every structure.

- **25 mph sustained or gusts to 35:** no work on open steel, roofs or swing stages.
- **Blowing sand:** goggles and respirator on, or come down.
- **Continuous siren:** clip in, make safe, climb down, go to shelter.
- Loose material at height gets tied down before you leave it.

HEAT AND HEIGHT STACK UP

A harness traps heat against your body. Steel decks and beams pass **158°F** in direct sun and burn skin in about a second. Heat dizziness at Deck 0 is a stumble. At Deck 12 it's a fall.

- Gloves and full coverage for any work on sun-baked steel. Kneel on a pad.
- Water and shade on the deck you're working on, not 12 decks down.
- Dizzy, cramping or confused? Clip in, sit down, call your buddy. Come down with help. (Module 3)
- Climbing counts as heavy work on the WBGT table.

STOP-WORK AUTHORITY

Every worker on this site can stop work when someone is working at an edge without protection, a guardrail is down, a hole is open, or there is no rescue plan. Stop it, make it safe, tell your crew lead.

WORKER ACKNOWLEDGEMENT

I know the 6-foot rule, I can inspect and fit my harness, I know how to figure fall clearance, and I will not climb without a rescue plan.

NAME	COLONIST ID	CREW / SECTOR	DATE

Adapted from Old Earth standards: OSHA 29 CFR 1926 Subpart M (1926.501 duty to have fall protection, 1926.502 guardrails, covers and personal fall arrest), Subpart L scaffolds (1926.451), Subpart X ladders (1926.1053), ANSI/ASSP Z359 fall protection code.